

Invasive earthworms alter forest soil microbiomes and nitrogen cycling

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ABSTRACT

Northern hardwood forests in formerly glaciated areas had been free of earthworms until exotic European earthworms were introduced by human activities. The invasion of exotic earthworms is known to dramatically alter soil physical, geochemical, and biological properties, but its impacts on soil microbiomes are still unclear. Here we show that the invasive earthworms alter soil microbiomes and ecosystem functioning, especially for nitrogen cycling. We collected soil samples at different depths from three sites across an active earthworm invasion chronosequence in a hardwood forest in Minnesota, USA. We analyzed the structures and the functional potentials of the soil microbiomes by using amplicon sequencing, high-throughput nitrogen cycle gene quantification (NiCE chip), and shotgun metagenomics. Both the levels of earthworm invasion and soil depth influenced the microbiome structures and the functional gene abundances. In the most recently and minimally invaded soils, *Nitrososphaera* and *Nitrospira* as well as the genes related to nitrification were more abundant than in the heavily invaded soils. By contrast, genes related to denitrification and nitrogen fixation were more abundant in the heavily invaded than the minimally invaded soils at various depths. Since denitrification can cause a nitrogen loss from the ecosystem, our results suggest the invasion of earthworms could influence the overall forest N cycling.

1. Introduction

Earthworms are well-known ecosystem engineers that shape soil structure and drive nutrient dynamics in soil ecosystem (Crooks, 2002). They feed on litter and soil, burrow horizontally and vertically through soils, and release fecal materials to mix nutrients in soils, altering soil porosity, bulk density, water infiltration, gas emission, nutrient mineralization, and plant productivity (Blouin et al., 2013).

Although earthworms are widely considered ubiquitous across the forest, grassland, agricultural, and garden ecosystems in the world, their global distribution is only beginning to be synthesized (Phillips et al., 2019). Glaciers and peri-glacial environments cleared out native earthworm populations from large areas in the northern USA and Canada as well as other Arctic areas in Eurasia during the last Ice Age (James and Hendrix, 2004). Since then, most of these areas had remained earthworm-free until European earthworm species were introduced by human activities (Hendrix et al., 2008).

The earthworm invasion is now widely regarded as a force that substantially alters physical, geochemical, and biological properties of

soils in northern hardwood forests (Bohlen et al., 2004b; Ferlian et al., 2018), and its ecosystem effects are believed to harm plant diversity (Craven et al., 2017) and be increasingly detrimental with ongoing changes in land uses and climates (Frelich et al., 2019). Invasive earthworms are known to mix the litter layer (O horizon) with underlying mineral soil, which results in creating A horizon at the expense of O horizon (Lyttle et al., 2011). Presumably coupled with the loss of O horizon, invasion of European earthworms is known to increase nitrate leaching in the formerly glaciated deciduous forests (Bohlen et al., 2004a). Denitrification enzyme activity was also higher in the forest soils with earthworms than in those without earthworms (Burtelow et al., 1998).

Earthworms can be divided into three ecological groups based on their feeding and burrowing behavior. Epigeic earthworms live and feed in the surface litter, while endogeic earthworms burrow mineral soils, ingesting decomposed matters and microbes together with minerals. Anecic earthworms, although living in deep vertical burrows, feed on leaf litter at the soil surface (Briones et al., 2005). The relative importance of these earthworm ecological groups, however, change as

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earthworm invasion progresses (Hale et al., 2005a; Wackett et al., 2018), indicating that ecosystem impacts of earthworm invasion are a strong function of invasion history and soil depth.

A limited number of studies suggest that the invasion of earthworms could alter soil microbial communities. For example, Dempsey et al. (2011) observed changes in soil microbial community composition based on the phospholipid fatty acid analyses (PLFA) in a northern hardwood forest in New York, USA. Although PLFA analysis provides quantitative information, it cannot provide community compositions at low taxonomic (e.g., genus) levels. Hoeffner et al. (2018) and de Menezes et al. (de Menezes et al., 2018) used terminal restriction fragment length polymorphism (T-RFLP) and high-throughput 16S rRNA gene amplicon sequencing, respectively, to analyze the effects of invasive earthworms on soil bacterial communities. However, these studies analyzed short-term impacts (e.g., 10–20 days (Hoeffner et al., 2018) and 17 weeks (de Menezes et al., 2018)) by using soil mesocosms. Field investigation is essential to analyze the longer-term impacts of earthworm invasion in natural soil environments.

Previously, we studied the impacts of earthworm invasion on the soil physicochemical properties in a northern hardwood forest in Minnesota, USA (Lyttle et al., 2011, 2015; Resner et al., 2015), which built on decades-long research on ecological processes and effects of earthworm invasion (Hale et al., 2005a, 2005b, 2006; Larson et al., 2010). At this site, a gradient of earthworm density and species composition was observed within a 190-m distance (Resner et al., 2015). The invasion of earthworms has drastically changed the cycling of carbon, nitrogen, and other inorganic nutrients in soils (Hale et al., 2005b; Lyttle et al., 2015; Resner et al., 2015). However, it remains unclear how soil microbial communities change in response to earthworm invasions. Since different ecological groups of earthworms occupy different stages of invasion, we hypothesized that the effect of earthworms on soil microbiomes varies by earthworm species and soil depth. In addition, since microbes play crucial roles in soil C and N cycling in forest ecosystems (Lladó et al., 2017), we hypothesized that the abundances of microbes important for C and N cycling have been changed by the invasion of earthworms.

Consequently, the objectives of this study were to (1) elucidate the impacts of earthworm invasion on bacterial, archeal, and fungal communities in natural forest soils at different depths, (2) clarify the relationships between the levels of earthworm invasion, microbial communities, and soil physicochemical properties, and (3) identify the C- and N-cycle related microbes/genes that are differentially abundant between minimally and heavily invaded soils.

2. Materials and methods

2.1. Soil sample collection

Soil samples were collected from a formerly glaciated northern hardwood forest near Leech Lake in north-central Minnesota, USA, on June 7, 2017 (Fig. S1). A detailed survey of soil physical properties and dendrochronology established that European earthworms arrived at the forest edge (recreational road) in the 1970s (Hale et al., 2005a; Larson et al., 2010) and have since invaded the forest for about 200 m (Lyttle et al., 2015; Resner et al., 2015). It has been well established that environmental variables including climate, vegetation, geology, and topography are consistent within the transect as described in the supplementary materials.

Based on the earthworm biomass, species composition, and the time since the arrival of European earthworms, we selected three sites along the transect. The heavily invaded site (H) (47°16′0.5″N, 94°23′55.1″W, 390 m ASL) was the nearest from the presumed source of invasion (i.e., a road) and had the largest earthworm biomass, dominated by anecic and endogeic species (Fig. S2). The minimally invaded site (M) (47°15′59.9″N, 94°23′45.9″W, 411 m ASL) was located ~210 m from site H and had the smallest earthworm biomass. Earthworms in the M site was dominated by epigeic earthworms (Fig. S2). The intermediate

site (I) (47°16′0.3″N, 94°23′48.4″W, 418 m ASL) was located between site H and site M (~150 m from site H). Although the total amount of earthworm biomass was similar between site H ($5.40 \pm 0.96 \text{ g/m}^2$) and site I ($4.50 \pm 0.68 \text{ g/m}^2$), their species compositions were different (Fig. S2) (Lyttle et al., 2015; Resner et al., 2015). Most notably, the population of anecic earthworms was much smaller in site I than in site H.

The soil type and plant vegetation were identical at these sites. The soil at these sites was mapped as the Warba soil series and classified as a fine-loamy, mixed, superactive, frigid Haplic Glossudalf (<https://websoilsurvey.sc.egov.usda.gov/>). However, organic matter contents were different most likely due to earthworm activities. Carbon content in the surface soil at the M site was 25.3% whereas those at the H and I sites were 7.4% and 11.4%, respectively (Lyttle et al., 2015). At the M site, a thick O horizon (>5 cm) and a thin A horizon (<2 cm) were observed, although the boundary between the O and A horizons was not always clear (Fig. S3). At the I site, the O horizon was <2 cm, whereas no clear O horizon was observed at the H site. At the H and I sites, however, A horizon was >5 cm (Resner et al., 2015). The most dominant tree species at this forest was sugar maples (*Acer sacharum*), but birch (*Betula papyrifera*, *Betula alleghaniensis*) and basswood (*Tilia americana*) also occurred.

After removing large undegraded leaves from the surface, three replicate soil cores (0–20 cm) were taken at each site by using a surface-disinfected soil probe (Fig. S3). The soil core samples were divided into six segments (0–2 cm, 2–4 cm, 4–6 cm, 6–8 cm, 8–10 cm, and 10–20 cm by depth) and placed in Whirl-Pak bags. A total of 54 soil samples were collected (three sites $\times$ three soil cores $\times$ six segments). Samples were kept on ice immediately after collection, and frozen with dry ice within 2 h of collection.

2.2. Soil physicochemical parameters

Ammonium, nitrate, nitrite, and moisture contents in the soil profiles at each site were measured in this study. Ammonium ion was extracted from soil (2 g) by shaking with 30 ml of 2 M KCl for 30 min. Nitrate and nitrite ions were extracted from soil (2 g) by shaking with 30 mL 0.01 M CaSO_4 for 15 min. Ammonium, nitrate, and nitrite contents in the soil extracts were measured using a Lachat QuikChem 8500 Flow Injection Analyzer in the Soil Testing and Research Analytical Laboratory at the University of Minnesota. Soil moisture content was measured by drying the samples at 105 °C for overnight. Soil pH, bulk densities, and carbon contents were measured for the samples collected in 2009 as described previously (Lyttle et al., 2015) and were used in this study.

2.3. DNA extraction, PCR, and amplicon sequencing

Total DNA was extracted from 0.25 g of each soil sample by using a DNeasy PowerSoil Kit (Qiagen) and QIAcube (Qiagen) according to the manufacturer's instructions. From these DNA samples, the V4 region of the 16S rRNA gene and the fungal internal transcribes spacer 2 (ITS2) region between 5.8S and 23S rRNA gene were amplified by using quantitative PCR (qPCR) with Meta_V4_515F and Meta_V4_806R primer set (Gohl et al., 2016) and 5.8SR_Nextera and ITS4_Nextera primer set (Castle et al., 2018), respectively. The qPCR mastermix was prepared using KAPA HiFidelity Hot Start Polymerase (Kapa Biosystems) as described previously (Gohl et al., 2016). The qPCR was done by using ABI 7900HT Real-Time PCR System (Applied Biosystem) with the following thermal conditions: 95 °C for 5 min, followed by 35 cycles of 98 °C for 20 s, 55 °C for 15 s, and 72 °C for 1 min, and one cycle of 72 °C for 5 min. Melting curve analysis and agarose gel electrophoresis were done to confirm the correct amplification of the PCR products.

For sequencing library preparation, the V4 region of the 16S rRNA gene and the fungal ITS2 region were amplified as described above, except that the number of PCR cycles was decreased to 25 based on the qPCR results. This number of PCR cycles was set to reach the log-linear

phase of the PCR kinetics to minimize PCR-dependent biases during amplification (Ishii et al., 2014). PCR products were purified and used to attach dual index tags and Illumina sequencing adaptors by PCR with Nextera kit as described previously in detail (Gohl et al., 2016; Castle et al., 2018). The pooled and size-selected amplicons were diluted to 8 pM in Illumina's HT1 buffer. Paired-end sequencing reactions were done using a MiSeq platform (Illumina) with V3 chemistry (300-bp read length) at the University of Minnesota Genomics Center (UMGC).

The paired-end raw sequence reads were quality-filtered, trimmed, and assembled using NINJA-SH17 (Al-Ghalith et al., 2018). The number of high-quality sequences per sample ranged from 28,079 to 284,434 and from 967 to 53,610 for the 16S rRNA gene and ITS2 region, respectively (Table S1). The sequences were clustered into operational taxonomic units (OTUs) at 97% sequence similarity by using NINJA-OPS (Al-Ghalith et al., 2016). Taxonomic assignments of the archaeal/bacterial and fungal OTUs were done using the Greengenes database version 97 (McDonald et al., 2012) and UNITE (Kõljalg et al., 2013) reference data sets, respectively. The resulting OTU tables with taxonomic information were used for statistical analyses (see below).

Based on the fungal OTU sequence data, fungal trophic modes and functional guilds were predicted by using the FUNGuild software (Nguyen et al., 2016). Only results with the confidence scores of "Probable" and "Highly Probable" were used for statistical analyses.

2.4. Nitrogen Cycle Evaluation (NiCE) chip

High-throughput microfluidic qPCR was used to quantify nitrogen cycle-associated genes (Nitrogen Cycle Evaluation [NiCE] chip) (Oshiki et al., 2018). Several assays were newly added to the NiCE chip system to increase the target coverage. A total of 43 qPCR assays were included, targeting the genes associated with nitrification, denitrification, dissimilatory nitrate reduction to ammonium (DNRA), anaerobic ammonium oxidation (anammox), and nitrogen fixation (Table S2).

The gBlock DNA fragments containing each of the target gene sequences were used as the standard DNA for the NiCE chip. When synthesis was difficult due to high GC contents, target gene fragments were PCR-amplified and cloned into a qPCR 2.1 plasmid vector (Thermo Fisher). The plasmids were linearized and purified as described previously (Ishii et al., 2013). All of the standard DNA fragments were mixed and diluted to make a serial dilution of 5×10^6 , 5×10^5 , 5×10^4 , 5×10^3 , 5×10^2 , 5×10^1 , and 5×10^0 copies/ μ L.

To increase the amount of the template DNA molecules, a pre-amplification (specific target amplification; STA) reaction was done prior to the NiCE chip run. The STA reaction is a multiplex PCR done with all primers with a small number of PCR cycles (Ishii et al., 2013). Both samples and standards were subjected to the STA reaction. The STA reaction mixture (8 μ L) included $1 \times$ TaqMan PreAmp Master Mix (Thermo Fisher), 0.05 μ M each primer, and 2 μ L of template DNA. The STA reaction was done using a Veriti Thermal Cycler (Applied Biosystems) with the following condition: 95 °C for 10 min, followed by 14 cycles at 95 °C for 15 s and 60 °C for 4 min. To remove unreacted primers, 0.5 μ L Exonuclease I (New England Biolabs) was added to each STA product and incubated at 37 °C for 30 min. The reaction was terminated by incubating the plate at 85 °C for 20 min. Reaction mixtures were then diluted to 40 μ L with TE buffer (10 mM Tris-HCl and 0.1 mM EDTA [pH 8]).

The NiCE chip was run using a DynamicArray 96.96 GE chip and a BioMark HD System (Fluidigm). The sample premix (8 μ L/sample) contained $1 \times$ SsoFast EvaGreen Supermix with low ROX (BioRad), $1 \times$ DNA Binding Dye (Fluidigm), and 3.6 μ L of Exo 1-treated and diluted STA product. The assay premix (8 μ L/assay) contained $1 \times$ Assay Loading Reagent (Fluidigm) and 5 μ M each forward and reverse primers. The sample premix and assay premix were dispensed onto the DynamicArray 96.96 GE chip and mixed by using an IFC controller HX instrument (Fluidigm), according to the manufacturer's instructions. After mixing, the reaction mixtures contained 0.5 μ M each forward and

reverse primers. qPCR was performed under the following conditions: 50 °C for 2 min, 70 °C for 30 min, 25 °C for 10 min, 95 °C for 10 min, and 40 cycles of 96 °C for 15 s, 50 °C for 30 s, and 72 °C for 1 min, followed by melting curve analysis from 50 °C to 95 °C at a rate of 1 °C/3 s. ROX was used as a passive dye.

The quantification cycle (C_q) was determined for each assay by using Real-Time PCR Analysis Software version 3.0.2 (Fluidigm). The standard curves were generated by linear regression analysis of the C_q values versus the known amounts of the standard DNA (log copies/ μ L) as described previously (Ishii et al., 2013). Quantities of target genes in the soil samples were calculated from the C_q values (with 2–3 replicates) using the standard curves. Limit of quantification (LOQ) was defined as the lowest concentration of the standards that were reliably detected (2.7–3.7 log copies/ μ L). Assays were excluded from the downstream analysis when >40% of samples were below LOQ. When <40% of samples were below LOQ, these samples were given $\frac{1}{2}$ LOQ values (Hites, 2019).

2.5. Shotgun metagenomic sequencing

DNA extracted from the surface soils (0–2 cm depth) at the three sites (H, I, and M) were also used for shotgun metagenomics. One of the replicates at the site I did not have enough amount of DNA, and therefore, was not used for metagenome analysis. As a result, eight samples were used for sequencing. Sequence libraries were created by using the Nextera XT kit (Illumina) with the mean library size of ~750 bp. Paired-end sequencing (150-bp read length) reaction was done using a NovaSeq platform (Illumina) at the UMG. To ensure sequence quality for downstream analysis, Shi7 was used to pre-process raw metagenomic sequence reads (Al-Ghalith et al., 2018). Shi7 removed the 3'- and 5'-adaptors from sequence reads, and trimmed them at a minimum quality score of 25 and a minimum length of 50 bp.

High-quality metagenomic sequence reads were mapped against the Greengenes and UNITE databases to identify bacterial/archaeal and fungal communities, respectively, by using bowtie2 which is implemented in the NINJA-OPS pipeline (Al-Ghalith et al., 2016). The resulting read count data were used for statistical analyses as described below.

To identify N cycle-related genes, we mapped the high-quality metagenomic sequence reads against the NCycDB, a comprehensive nitrogen cycle protein sequence database (Tu et al., 2018), by using DIAMOND (Buchfink et al., 2015) with a minimum sequence identity of 70%, a minimum query coverage of 75%, and an E-value of $<10^{-5}$. To identify fungal denitrification-associated genes, we used the reference sequences reported by Higgins et al. (2018) for the read mapping. Description of the top-hit sequence (i.e., aligned with the lowest E-value) was extracted for each mapped read, grouped according to their potential functions, and used to create a read count table. The number of reads mapped to each of the sequence groups (i.e., gene functions) was normalized by the total number of high-quality sequence reads for each sample and used for statistical analyses.

High-quality metagenomic sequence reads were also mapped against the Greengenes and UNITE databases to identify bacterial/archaeal and fungal communities, respectively, by using bowtie2 which is implemented in the NINJA-OPS pipeline (Al-Ghalith et al., 2016). The resulting read count data were used for statistical analyses.

2.6. Statistical analyses

Statistical significances in the quantitative data obtained in this study were tested with the Kruskal-Wallis rank-sum test by using R version 4.0.2 (<https://www.r-project.org/>). Microbial community structures were analyzed by using R with *vegan* (Oksanen et al., 2019), *phyloseq* (McMurdie and Holmes, 2013), and *DESeq2* packages (Love et al., 2014). The numbers of sequences were normalized (i.e., rarefied) at the smallest number of sequences by random subsampling for the

diversity analyses. The subsampled sequences provided sufficient resolution of the microbial communities, as indicated by Good's coverage ranging from 0.952 to 0.992. Principal coordinates analysis (PCoA) with Bray-Curtis distance matrices were used to visualize the dissimilarities in microbial communities among sites (i.e., the levels of earthworm invasion) and soil depth. Differences in microbial community structures were tested using permutational multivariate analysis of variance (PERMANOVA).

Canonical analysis of principal coordinates (CAP) was done using Bray-Curtis distance matrices to identify environmental variables associated with the patterns in microbial communities (Anderson and Willis, 2003). Environmental variables that were correlated with other variables at the Spearman's ρ values of >0.80 were removed from the CAP analysis. Multicollinearity among the environmental variables was also identified by calculating variance inflation factors (VIF). Variables with a VIF of >10 were removed from the model in the CAP analysis. Furthermore, variables that were not significant ($p > 0.05$) by PERMANOVA were removed from the model.

Taxa or genes that increased or decreased their relative abundance after earthworm invasion were identified by Spearman's rank correlation analysis between gene abundance and earthworm biomass, which was the largest in the heavily invaded soil and the smallest in the minimally invaded soils (Fig. S2). In addition, differentially abundant taxa across samples were identified using DESeq2 with $\alpha = 0.01$.

Generalized linear models were used to identify the environmental variables such as earthworm biomass, soil depth, and soil physico-chemical parameters that can explain the quantities of N cycle genes. Multimodel inference was done using R with *glmulti* package (Calcagno and de Mazancourt, 2010).

2.7. Nucleotide sequence accession numbers

The 16S rRNA gene and fungal ITS2 amplicon sequences as well as the shotgun metagenomics sequence reads were deposited to the GenBank database under the BioProject number PRJNA504043. The SRA accession numbers are available in Tables S1 and S3.

3. Results

3.1. Soil ammonium and nitrate concentrations at different depth

The soil ammonium and nitrate concentrations decreased by depth ($p < 0.05$ by Kruskal-Wallis test) at all sites (Fig. 1). A similar trend was seen when concentrations were expressed per m^3 soil (Fig. S4). The

ammonium concentrations in the deeper soil (4–8 cm depth) were higher in the earthworm-invaded site (site H and I) than in the minimally invaded site (site M) most likely due to the soil mixing by earthworms in the H and I sites. Similarly, higher nitrate concentrations were seen in the 6–10 cm soils in site I than those in site M ($p < 0.05$). Nitrate was not detected in the deep soils in site H.

3.2. Abundance of microbes in soils

While abundances of archaea/bacteria (Fig. 2A) and fungi (Fig. 2B) were not significantly different by the levels of earthworm invasion (i.e., sites), they were significantly different by depth ($p < 0.01$ by Kruskal-Wallis test). Both archaea/bacteria and fungi were most abundant in near-surface soils. Soil depth was negatively correlated with the copy numbers of the 16S rRNA gene (Spearman's $\rho = -0.77$, $p < 0.01$) and the ITS2 region (Spearman's $\rho = -0.60$, $p < 0.05$). Although differences in the abundance of archaea/bacteria between the surface soil (0–2 cm) and soils at 10–20 cm were similar by site, those between the surface soil and soils at 8–10 cm depth was larger in the minimally invaded soil (site M) than heavily invaded soil (site H) ($p < 0.05$ by Kruskal-Wallis test) (Table S4). The same trend was also seen for the differences in the fungal abundances between the surface soil (0–2 cm) and the soils at 8–10 cm (Table S4). In addition, the effect of soil depth on microbial abundances was also seen when abundances were normalized by soil C content (Fig. S5).

3.3. Alpha diversity measures

Species richness was inferred by the numbers of observed OTUs for archaea/bacteria (Fig. 2C) and fungi (Fig. 2D). The numbers of archaeal/bacterial and fungal OTUs decreased by depth ($p < 0.01$ and < 0.05 , respectively, by Kruskal-Wallis test), but not by the levels of earthworm invasion. Soil samples collected at 10–20 cm had the smallest number of OTUs for both archaea/bacteria and fungi, and the values were similar across sites. Differences in the numbers of observed OTUs for archaea/bacteria between the surface soil (0–2 cm) and the soils at 8–10 cm depth was larger in the site M than the sites H and I (Table S5).

Shannon diversity index values calculated based on the archaeal/bacterial 16S rRNA gene sequences were also significantly different by soil depth ($p < 0.01$ by Kruskal-Wallis test) (Fig. 2E), whereas those calculated based on the fungal ITS2 region were not (Fig. 2F). Shannon index values for archaea/bacteria were the smallest in the soil samples collected at the 10–20 cm samples, and these values were similar across

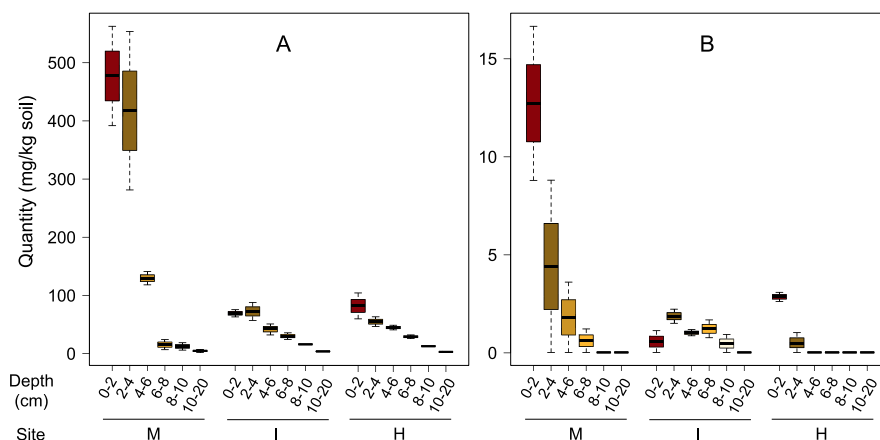


Fig. 1. Quantities of (A) ammonium and (B) nitrate in the soil samples (in mg/kg soil). Legend: M, the site with minimal invasion of earthworms; I, the site with an intermediate invasion of earthworms; and H, the site with a heavy invasion of earthworms.

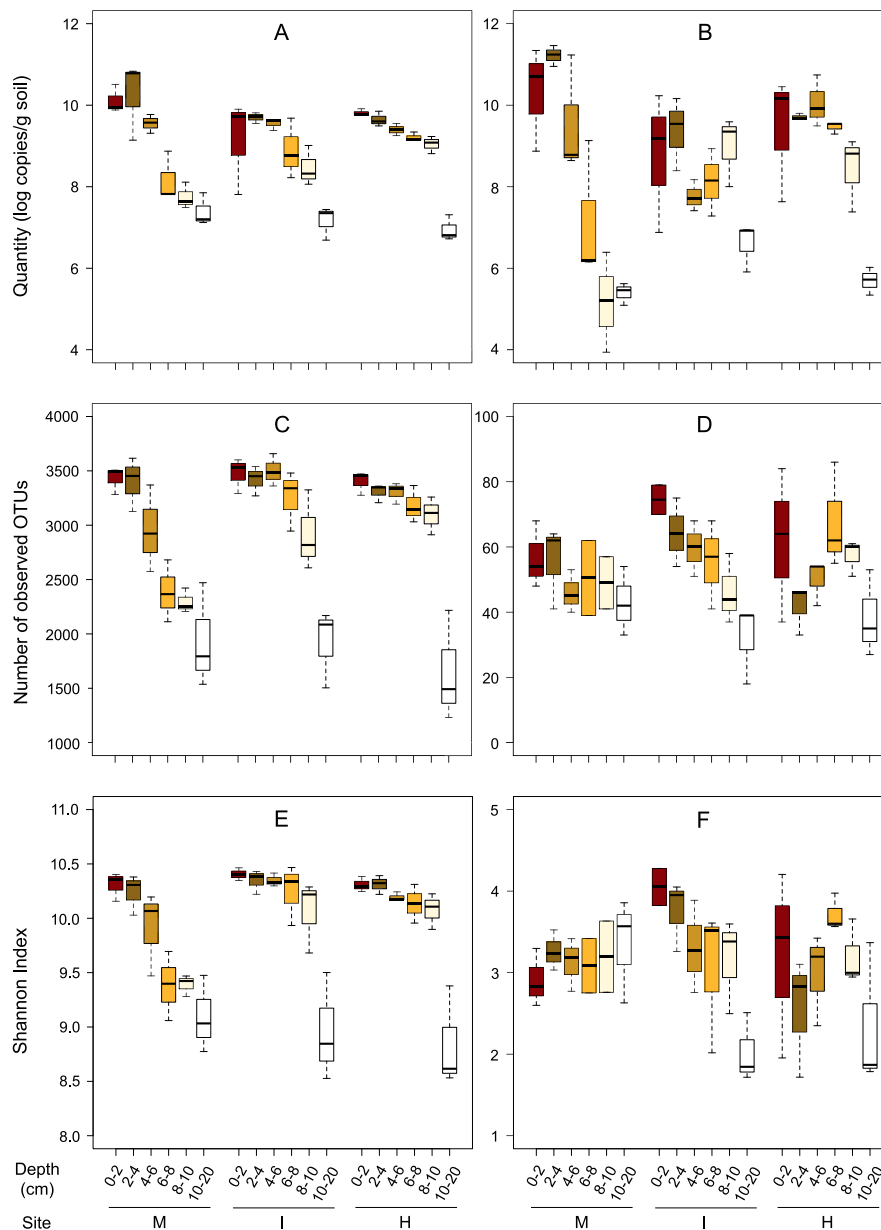


Fig. 2. Abundance and diversities of microbes in the soil samples. Quantities of (A) 16S rRNA gene and (B) fungal ITS2 region measured by qPCR. Numbers of observed OTUs for (C) archaea/bacteria and (D) fungi. Shannon diversity index values calculated based on (E) the archaeal/bacterial 16S rRNA gene and (F) fungal ITS2 region sequences. Legend: M, the site with minimal invasion of earthworms; I, the site with an intermediate invasion of earthworms; and H, the site with a heavy invasion of earthworms.

sites. Shannon diversity index values were not significantly different by the levels of earthworm invasion for both archaea/bacteria and fungi, although differences in the Shannon index values for archaea/bacteria between the surface soil (0–2 cm) and soils at 8–10 cm depth was greater in the site M than the sites H and I (Table S5).

3.4. Patterns in soil microbial communities

Principal coordinates analysis (PCoA) showed the grouping of soil archaeal/bacterial communities by the levels of earthworm invasions (site M vs. site I and H) and soil depth (Fig. 3A). Community dissimilarities among different levels of earthworm invasions and soil depths were supported by PERMANOVA ($p < 0.01$). The archaeal/bacterial communities in the near-surface soil (0–2 cm) and those in the deepest soil (10–20 cm) were most distinctly plotted to each other. Distances between the archaeal/bacterial communities in the soils with minimum earthworm invasion (site M) and those in the soils with intermediate and heavy invasions (site I and H, respectively) were larger in the shallow

soils (0–10 cm depths) than the deep soils (10–20 cm). Archaeal/bacterial communities in the deep soils overlapped each other, indicating that their community structures were similar to each other.

Fungal communities in the soils were also different by the levels of earthworm invasion ($p < 0.01$ by PERMANOVA). Fungal communities in the soils with minimum earthworm invasion were more closely related to each other than those in the soils with intermediate and heavy invasions on the PCoA plot (Fig. 3B). In contrast to archaeal/bacterial communities, fungal communities were not different by soil depth ($p = 0.38$ by PERMANOVA).

3.5. Taxonomic composition

Major archaeal and bacterial phyla identified in this study include *Acidobacteria*, *Actinobacteria*, *Bacteroidetes*, *Crenarchaeota*, *Nitrospirae*, *Planctomycetes*, *Proteobacteria*, and *Verrucomicrobia* (Fig. 4A). Relative abundances of these phyla were similar among the soils with different levels of earthworm invasions, except for *Crenarchaeota* and *Nitrospirae*.

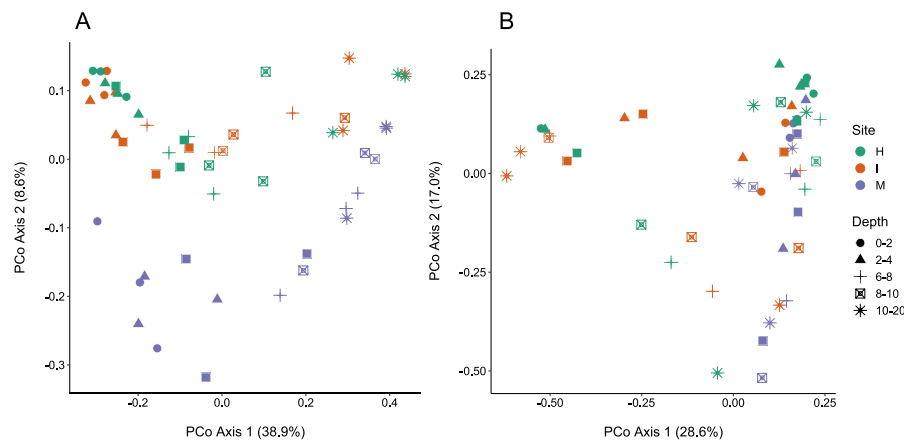


Fig. 3. Principal coordinate analysis (PCoA) plots showing the Bray-Curtis dissimilarities among (A) archaeal/bacterial communities and (B) fungal communities. Each community is labeled with the site (M, the site with minimal invasion of earthworms; I, the site with an intermediate invasion of earthworms; and H, the site with a heavy invasion of earthworms) and soil depth (0–2, 2–4, 4–6, 6–8, 8–10, and 10–20 cm from the surface). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

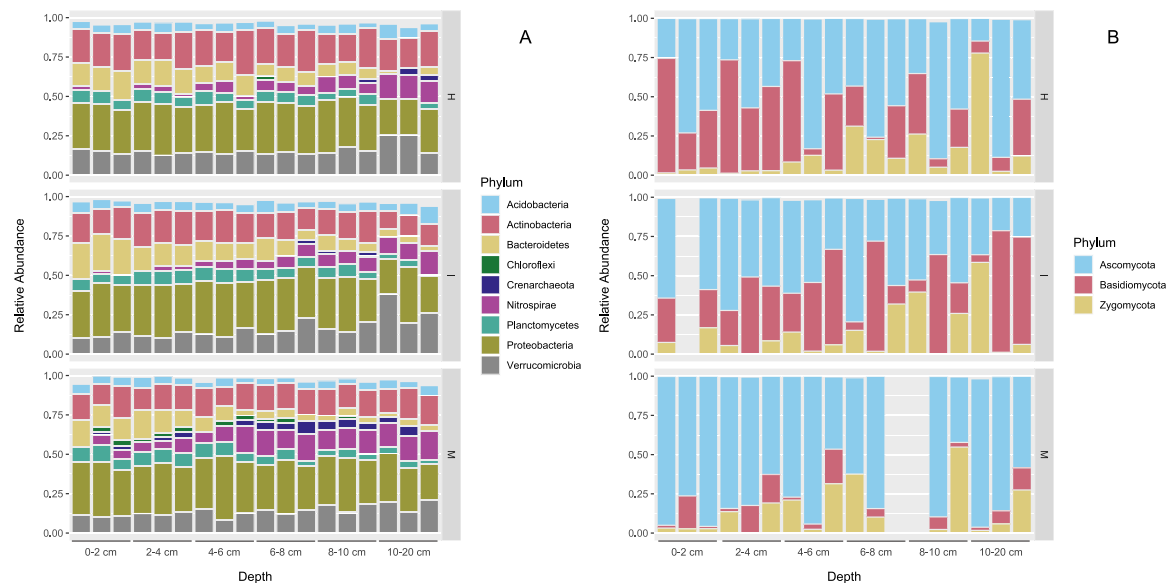


Fig. 4. Relative abundance of (A) archaeal/bacterial phyla and (B) fungal phyla in the soil samples as assessed by the 16S rRNA gene and ITS2 sequencing analyses. Legend: M, the site with minimal invasion of earthworms; I, the site with an intermediate invasion of earthworms; and H, the site with a heavy invasion of earthworms. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Relative abundances of *Crenarchaeota* and *Nitrospirae* were the largest in the soil with minimal invasion of earthworms (site M) ($p < 0.01$ by Kruskal-Wallis test). Similar results were obtained by the 16S rRNA gene amplicon sequencing and the shotgun metagenomics (Fig. S6).

Ascomycota, *Basidiomycota*, and *Zygomycota* were the major fungal phyla identified in this study (Fig. 4B). Relative abundances of *Basidiomycota* were larger in the soils with more earthworm invasions (site I and H) than those with minimal invasions earthworm invasions ($p < 0.01$ by Kruskal-Wallis test).

3.6. Microbial taxa responsive to earthworm invasion

Spearman's rank correlation analysis identified the taxa that increased or decreased their relative abundance after the invasion of earthworms (Table S6). The same taxa were also identified as differentially abundant among the sites by the DESeq2 program. Taxa that showed positive correlations with the levels of earthworm invasion included the genus *Mycobacterium* ($p = 0.56$, $p < 0.01$) in the phylum *Actinobacteria*. Taxa that showed negative correlations with the levels of

earthworm invasion included the genus *Nitrososphaera* ($p = -0.60$, $p < 0.01$) in the phylum *Crenarchaeota*, the genus *Nitrospira* ($p = -0.50$, $p < 0.01$) in the phylum *Nitrospirae*, and the fungal order *Helotiales* ($p = -0.57$, $p < 0.01$) in the phylum *Ascomycota*.

3.7. Quantities of the N cycle-associated genes

Some members of the genera *Nitrososphaera* and *Nitrospirae* play important roles in the N cycle, namely ammonia oxidation and nitrite oxidation, respectively. The decrease in their relative abundances in the earthworm-invaded soils could influence the overall N cycling in the soils. To clarify this, we used a high-throughput N-cycle gene quantification tool called the NiCE chip. Of the 43 assays included in the NiCE chip, 34 assays went successfully; i.e., standard curves were generated with $r^2 > 0.95$ (Table S2). Of those, 18 assays showed positive results in >60% of the samples. These 18 assays targeted genes for nitrification (*amoA* and *nirX*), denitrification (*napA*, *nirK*, *nirS*, *norB*, *nosZ*), and nitrogen fixation (*nifH*). In general, denitrification-related genes were more abundant than nitrification-related genes (Fig. 5A). Many of the

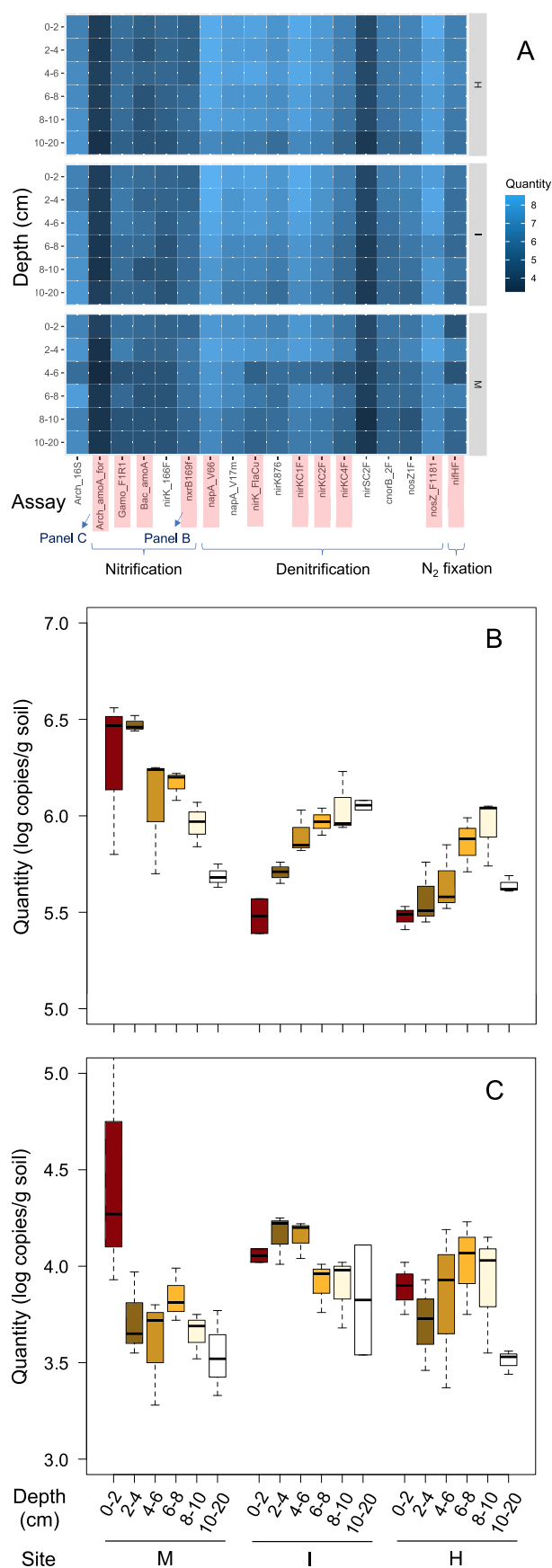


Fig. 5. Quantities of N cycle-related genes in the soil samples. (A) Pseudo-heatmap showing the Nitrogen Cycle Evaluation (NiCE) chip results. Assays producing positives in >60% of samples are shown. Assay names shaded with red indicate that the gene quantities measured by these assays were different by site by Kruskal-Wallis test ($p < 0.05$). (B) Quantities of *nitrB* of *Nitrospira*, based on the *nitrB169f* assay. (C) Quantities of archaeal *amoA* based on the Arch-*amoA* for assay. Other selected gene quantities (*nirK* and *nifH*) are shown in Fig. S7. Legend: M, the site with minimal invasion of earthworms; I, the site with an intermediate invasion of earthworms; and H, the site with a heavy invasion of earthworms. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

denitrification- and nitrogen fixation-related genes were also more abundant in the soils at various depths with heavy earthworm invasions than the soils with minimal invasions ($p < 0.05$ by Kruskal-Wallis test) (Fig. S7).

By contrast, the quantities of *nitrB* of *Nitrospira* (measured by the *nitrB169f* assay) were significantly greater in the soils with minimal invasion of earthworms than those heavily invaded by earthworms ($p < 0.01$ by Kruskal-Wallis test) (Fig. 5B), in agreement with the changes in the relative abundance of *Nitrospira* measured by 16S rRNA gene amplicon sequencing. Interestingly, while quantities of *Nitrospira nitrB* decreased by depth in the samples with minimal invasion of earthworms ($p < 0.01$ by Kruskal-Wallis test), those in the samples with heavy/intermediate invasions increased by depth ($p < 0.01$ by Kruskal-Wallis test).

Quantities of archaeal *amoA* were not significantly different among the samples with different levels of earthworm invasion (Fig. 5C). In minimally invaded soils, archaeal *amoA* levels were significantly greater in the surface 0–2 cm soils than the soils collected at 10–20 cm depth ($p < 0.05$ by Kruskal-Wallis and Mann's *post hoc* test).

Generalized linear models were used to identify the environmental variables important for the quantities of the denitrification gene (represented by *nirK*), nitrogen fixation gene (*nifH*), *Nitrospira nitrB*, and archaeal *amoA*. Earthworm density and soil depth were the variables most commonly included in the models (Table S7). Other important variables identified included soil bulk density (for *Nitrospira nitrB* and archaeal *amoA*), ammonium content (for *Nitrospira nitrB*), nitrate content (archaeal *amoA* and *nifH*), and pH (for *nirK*) (Table S7).

3.8. Environmental variables associated with the patterns in microbial communities

Canonical analysis of principal coordinates (CAP) was used to clarify the relationship between environmental variables and the patterns in microbial communities. To select environmental variables for the CAP analysis, we first did a correlation analysis (Fig. S8). Most denitrification genes correlated with each other at Spearman's $\rho > 0.8$. Assay *nirK*-*FlaCu* was selected as the representative assay for denitrification because this assay produced quantitative values for all samples. Based on the correlation analysis, 13 variables were selected and used for the CAP analysis. After the VIF analysis and PERMANOVA, seven and five variables remained in the final CAP models for archaeal/bacterial and fungal communities, respectively (Fig. 6). Earthworm biomass, ammonium concentration, soil bulk density, and the quantities of denitrification and nitrogen fixation genes (*nirK*-*FlaCu* and *nifH* assays, respectively) were commonly identified as the variables significantly associated with the patterns in both archaeal/bacterial and fungal communities ($p < 0.05$ by PERMANOVA). For archaeal/bacterial communities, additional two variables, quantities of nitrification genes (*Arch_amoA* for and *nitrB169f*) were also identified.

3.9. Functional potentials of the soil microbial communities

To further assess the functional potential for N cycling in the soil microbial communities including those not of our original target (e.g., N

(caption on next column)

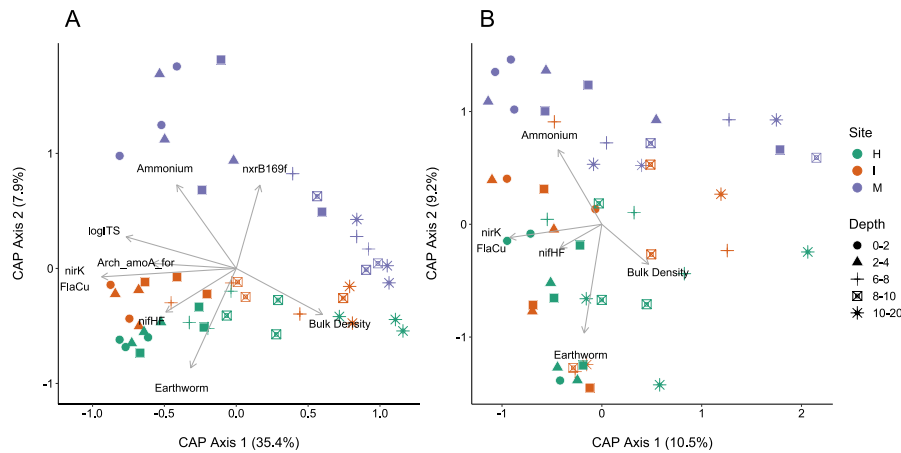


Fig. 6. Canonical analysis of principal coordinates (CAP) plots showing the associations between environmental variables and the patterns in (A) archaeal/bacterial communities and (B) fungal communities. Microbial communities were analyzed using Bray-Curtis distance matrices. All environmental variables shown in the plots had significant effects ($p < 0.05$) on the microbial community patterns based on PERMANOVA. Each community is labeled with the site (M, the site with minimal invasion of earthworms; I, the site with an intermediate invasion of earthworms; and H, the site with a heavy invasion of earthworms) and soil depth (0–2, 2–4, 4–6, 6–8, 8–10, and 10–20 cm from the surface). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

assimilation and mineralization), we used the shotgun metagenomics approach. Surface soils (0–2 cm) were used for the metagenomic analysis because microbiomes in these soils were most significantly different between sites H, I, and M. For each sample, 26.3–39.8 M reads were obtained and used for the analysis (Table S3). Similar to the NiCE chip results, the relative abundances of the genes responsible for denitrification (*nar*, *nap*, *nor*, and *nos*) and nitrogen fixation (*nif*) were positively correlated with the levels of earthworm invasion (Spearman's $\rho > 0.75$, $p < 0.05$) (Fig. 7). The relative abundances of bacterial/archaeal nitrite reductase genes (*nirK* and *nirS*) and fungal nitrite reductase gene (fungal *nirK*) were also positively correlated with the levels of earthworm invasion (Spearman's $\rho = 0.57$ and 0.69 , respectively), although they were not statistically significant ($p = 0.143$ and 0.057 , respectively) (Table S8). By contrast, the abundance of bacterial and archaeal ammonia monooxygenase genes for nitrification was negatively associated with the earthworm invasion levels (Spearman's $\rho < -0.80$, $p < 0.05$). The relative abundance of nitrite oxidoreductase gene (*nrx*) was

also negatively yet insignificantly correlated with the levels of earthworm invasion (Spearman's $\rho = -0.57$, $p = 0.143$).

Based on the metagenomic data, the genes related to N assimilation such as glutamine synthetase and asparagine synthase as well as those related to N mineralization such as glutamate dehydrogenase were more abundant than the nitrification and denitrification-related genes (Table S8). The relative abundance of glutamate dehydrogenase was positively correlated with the levels of earthworm invasion (Spearman's $\rho = 0.76$, $p < 0.05$), although other genes were not (Table S8).

To assess the potential functions of the soil fungal communities, we used the FUNGuild approach by using the ITS amplicon sequencing data obtained from all depth-increment samples at all sites as the input. The relative abundance of symbiotrophs (e.g., ectomycorrhizae) was positively correlated with the levels of earthworm invasion (Spearman's $\rho > 0.46$, $p < 0.01$); whereas those of saprotrophs and pathotrophs were negatively correlated with the levels of earthworm invasion (Spearman's $\rho < -0.41$, $p < 0.01$) (Table S8).

4. Discussion

While the invasion of earthworms itself did not influence the abundances of archaea/bacteria and fungi as well as the α diversity measures, soil depths had significant impacts on the quantities and α diversities of soil microbiota. Effects of soil depths on microbial communities have been well documented in temperate forest soils (Eilers et al., 2012; Tang et al., 2018). Impacts of soil depth were greater in the minimally invaded soil (site M) than more earthworm-invaded soils (sites H and I), as we found larger differences in the quantities and diversities of soil archaeal/bacterial populations between the surface (0–2 cm) and deep (8–10 cm) soils in minimally invaded soil than in heavily invaded soils. This is likely due to the mixing effects by soil-dwelling earthworms such as endogeic and anecic earthworms that were abundantly present in sites H and I. These earthworms vertically mix organic materials with minerals to form A horizon (Lyttle et al., 2015; Resner et al., 2015). Interestingly, differences in the quantities and α diversities of soil microbiota between the surface (0–2 cm) and the deepest (10–20 cm) soils were not significantly different by the level of earthworm invasion, suggesting that the impacts of earthworm invasion on soil microbial communities were minimal in the soils at 10–20 cm depth. This agrees with our field observations that A horizons rarely exceed 10 cm depth (Resner et al., 2015).

Both soil depth and the level of earthworm invasion had significant impacts on β diversity. The impact of earthworm invasion on β diversity of soil archaeal/bacterial communities was greater in shallow soils (0–10 cm) than in deep soils (10–20 cm). Indeed, archaeal/bacterial

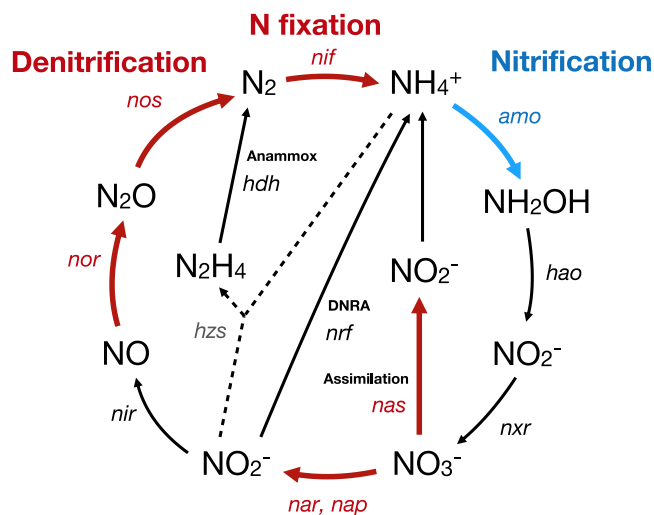


Fig. 7. Nitrogen cycling in the forest soil influenced by earthworm invasion. Red and blue arrows indicate the genes that increased and decreased their relative abundances by earthworm invasion, respectively, based on Spearman's correlation analysis of the shotgun metagenomics reads ($p < 0.05$). Black solid arrows indicate the genes that did not change their abundance by earthworm invasion. Black dashed arrows indicate the genes that were not detected. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

communities in the deep soils were similar to each other regardless of the levels of earthworm invasion. Although some anecic earthworms are known to dig deep burrows of up to 2 m (Bohlen et al., 2004b), anecic *L. terrestris* have never been observed below A horizons at our earthworm invasion transect presumably because of the dense clay-rich B horizon (Resner et al., 2015). This observation agrees with our results that the invasion of earthworms had limited influence on the archaeal/bacterial community structures in the soils at 10–20 cm.

Acidobacteria, *Actinobacteria*, *Bacteroidetes*, *Planctomycetes*, *Proteobacteria*, and *Verrucomicrobia* occupied relatively large proportions of the archaeal and bacterial populations. Similar to this study, these phyla have been commonly detected in soils of temperate deciduous forests (López-Mondéjar et al., 2015; Isobe et al., 2018). Relatively large proportions of *Crenarchaeota* and *Nitrospirae* were also detected in this study, especially in the minimally invaded soils. These phyla are not frequently detected in other forest soils (e.g., (López-Mondéjar et al., 2015; Isobe et al., 2018)), most of which presumably contain earthworms. *Nitrososphaera* spp. in the phylum *Crenarchaeota* and *Nitrospira* spp. in the phylum *Nitrospirae* were more abundant in the minimally invaded soils than the other soils. These microbes may play important roles in nitrification (Tourna et al., 2011; Wang et al., 2019; Li et al., 2020; Osburn and Barrett, 2020). These microbes as well as nitrification functional genes were more abundant in the minimally invaded soils than more earthworm-invaded soils, indicating that nitrification could occur more actively in the minimally invaded soils. This is most likely due to the higher availability of nitrification substrate (i.e., NH_4^+) in these soils.

Interestingly, the vertical distribution of *Nitrospira* was different between the minimally invaded soils and more earthworm-invaded soils. *Nitrospira* spp. were more abundant near the surface of minimally invaded soils, probably because substrate for nitrification (ammonium and nitrite) were more available near the surface (O horizon). By contrast, *Nitrospira* spp. were more abundant in deeper soils of earthworm-invaded sites, probably because of the mixing soil microbes by earthworms and/or the leaching of nitrite downward. The GLM analysis also shows the importance of soil depth, earthworm biomass, soil bulk density, and the concentrations of nitrification-related compounds (ammonium and nitrate) to explain the quantity of archaeal *amoA* and *Nitrospira nxrB*. Earthworm activity can increase soil bulk density (Frelich et al., 2019); therefore, these results suggest that the activity of invasive earthworms can influence the abundance of AOA and *Nitrospira* spp. in the soil at various depths.

While the relative abundance of *Nitrososphaera* spp. and *Nitrospira* spp. decreased by earthworm invasion, that of *Mycobacterium* (*Actinobacteria*) increased. Specific bacteria can be enriched in the earthworm guts, in which C, N, and other nutrients are more abundant than in soils (Medina-Sauza et al., 2019; Sun et al., 2020). Indeed, *Mycobacterium* was frequently isolated from the guts of anecic earthworms (*L. terrestris*) (Fischer et al., 2003) and was also identified as one of the bacterial taxa residing in earthworm gut walls of endogeic earthworms (*Aporrectodea* spp.) (Thakuria et al., 2010). In addition, *Mycobacterium* was present at a significantly larger proportion in the guts of epi-endogeic earthworms (*L. rubellus*) than in soils (Pass et al., 2015). These results collectively suggest that earthworms likely enriched *Mycobacterium* in their guts and contributed to the increased abundance of *Mycobacterium* in earthworm-invaded soils.

Metagenomic analysis revealed that archaeal *amoA* was more abundant in the minimally invaded soils than in the other soils, further supporting that archaea including *Nitrososphaera* spp. play ammonia oxidation in the minimally invaded soils. The qPCR analysis (i.e., NiCE chip), however, did not show a significant difference in the archaeal *amoA* abundance among the soils. Similarly, a discrepancy was also observed for the abundance of *nxrB* of *Nitrospira*: while NiCE chip results showed a significant difference in the abundance of *nxrB* of *Nitrospira* among the soils, no difference was detected by metagenomics. The discrepant results obtained by the two approaches may be due, in part,

to (1) the large variations seen in the qPCR results, which was likely caused by the relatively low abundance of archaeal *amoA* and *Nitrospira nxrB* in the samples, (2) biases caused by the PCR primers (i.e., not all target genes are necessarily amplified by qPCR), or (3) the difference in the quantitative nature of the methods: while qPCR can provide absolute quantification (copies/g soil), metagenomics approach can provide only relative quantification (counts per million reads) in this study (Ishii, 2020).

Both metagenomics and the NiCE chip analyses showed that many of the denitrification-related genes were more abundant in the earthworm-invaded soils than in the soils with minimal invasions. This was observed not only in the surface 0–2 cm soils but also in the deeper soils as shown by the NiCE chip results. This may appear contradictory to the fact that the concentration of denitrification substrate (i.e., nitrate) was higher in the minimally invaded soil than the heavily invaded soils as seen in this study as well as Hale et al. (2005b). However, it is probable that nitrate may have been consumed by denitrifiers in the earthworm-invaded soils, and therefore, became low compared to that in the minimally invaded soils. Similar to this study, greater denitrification activities were observed in earthworm-invaded forest soils than non-invaded soils (Scheu and Parkinson, 1994; Burtelow et al., 1998). Higher denitrification activity was also reported in earthworm excreta (i.e., casts) than in the surrounding soils (Svensson et al., 1986). The larger abundance of denitrifying organisms in earthworm-invaded soils might be due, in part, to the enrichment of denitrifiers in earthworm intestines and/or the presence of more anoxic areas in soils. Earthworm intestines are known to promote denitrification, most likely due to the presence of anoxic area and readily assimilable carbon (i.e., mucus) (Karsten and Drake, 1997; Horn et al., 2003; Medina-Sauza et al., 2019; Sun et al., 2020). Earthworm-invaded soils could also have more anoxic micro-sites than the minimally invaded soils. Along our earthworm invasion chronosequence, both soil bulk density and the size and strength of soil aggregation are positively related with the level of earthworm invasion (Hale et al., 2005b; Lyttle et al., 2015), which is likely to offer more anoxic micro-sites that favor denitrification in the heavily invaded soils.

In addition to denitrification-related genes, the genes related to nitrogen fixation (*nif*) were also enriched in the earthworm-invaded soils at various depth in this study. The impact of earthworm invasion in nitrogen fixation is not well known; however, since total N content can decrease by earthworm-associated activities (Eisenhauer et al., 2007), the soil environment may become N-limited, which provides a selective advantage for N-fixing microbes.

Regarding the soil fungal communities, the relative abundance of *Basidiomycota* increased by earthworm invasion at all depths. This might be related to the increased abundance of symbiotrophs identified by the FUNGuild analysis because some fungi in *Basidiomycota* are known as being ectomycorrhizae and symbiotically associate with trees (Martin et al., 2016). Ectomycorrhizae receive C mostly from their host plants instead of degrading complex organic matter in soils (Martin et al., 2016). By contrast, the relative abundance of the order *Helotiales* (*Ascomycota*) decreased by the invasion of earthworms, which may be related to the decrease of the saprotrophs in the earthworm-rich soils. Most members of *Helotiales* live as soil saprophytes and degrade dead woods or other organic matter (Tedersoo et al., 2009). Similar to this study, Dempsey et al. (2011) detected higher and lower levels of mycorrhizae and saprophytic fungi, respectively, in the earthworm-invaded soils than in the earthworm-free soils in a northern hardwood forest in New York, USA, by using the PLFA analysis. Collectively, these results suggest that soil fungal communities would shift from saprophytic to symbiotrophic communities by the invasion of earthworms, most likely due to the observed decrease in the organic C contents after earthworm invasions.

5. Conclusions

In conclusion, this study clearly shows that the invasion of earthworms alters soil microbial communities and ecosystem functioning. Earthworms mix soils, change soil physical structures and redox environments, decrease the levels of C, N, and other nutrients in soils, secrete mucus, and enrich specific microbes in their guts, all of which can influence the microbial activities in soils. The most notable changes that earthworm invasion causes include the shift in the soil N cycling. Our results suggest that nitrification occurs more actively in forest soils before the invasion of earthworms, for which AOA and *Nitrospira* play key roles. After earthworm invasion, denitrification would become more active, which may cause N-limited conditions and the increased importance of the N fixing populations. Since DNA-based analysis alone cannot verify the active occurrence of these processes, further research is needed to quantify and compare the activity of nitrification, denitrification, and N fixation between soils with different levels of earthworm invasion. In addition, future research should include more replicates both in terms of the number of earthworm invasion transects and the number of soil cores at each site to further improve the statistical performance.

The invasion of earthworms is ongoing. They are actively spreading in formerly glaciated areas such as Alaska (Saltmarsh et al., 2016) and northern Scandinavia (Wackett et al., 2018) in addition to Minnesota. To better understand the impacts of “global worming” on the ecology of soil systems, future research is necessary including the analysis of soil microbiomes at non-invaded and recently invaded soils on a global scale.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.soilbio.2022.108724>.

References

- Al-Ghalith, G.A., Hillmann, B., Ang, K., Shields-Cutler, R., Knights, D., 2018. SHI7 is a self-learning pipeline for multipurpose short-read DNA quality control. *mSystems* 3, e00202, 00217.
- Al-Ghalith, G.A., Montassier, E., Ward, H.N., Knights, D., 2016. NINJA-OPS: fast accurate marker gene alignment using concatenated ribosomes. *PLoS Computational Biology* 12, e1004658.
- Anderson, M.J., Willis, T.J., 2003. Canonical analysis of principal coordinates: a useful method of constrained ordination for ecology. *Ecology* 84, 511–525.
- Blouin, M., Hodson, M.E., Delgado, E.A., Baker, G., Brussaard, L., Butt, K.R., Dai, J., Dendooven, L., Peres, G., Tondoh, J.E., Cluzeau, D., Brun, J.-J., 2013. A review of earthworm impact on soil function and ecosystem services. *European Journal of Soil Science* 64, 161–182.
- Bohlen, P.J., Pelletier, D.M., Groffman, P.M., Fahey, T.J., Fisk, M.C., 2004a. Influence of earthworm invasion on redistribution and retention of soil carbon and nitrogen in northern temperate forests. *Ecosystems* 7, 13–27.
- Bohlen, P.J., Scheu, S., Hale, C.M., McLean, M.A., Migge, S., Groffman, P.M., Parkinson, D., 2004b. Non-native invasive earthworms as agents of change in northern temperate forests. *Frontiers in Ecology and the Environment* 2, 427–435.

- Briones, M.J.I., Garnett, M.H., Pearce, T.G., 2005. Earthworm ecological groupings based on ^{14}C analysis. *Soil Biology and Biochemistry* 37, 2145–2149.
- Buchfink, B., Xie, C., Huson, D.H., 2015. Fast and sensitive protein alignment using DIAMOND. *Nature Methods* 12, 59–60.
- Burtelow, A.E., Bohlen, P.J., Groffman, P.M., 1998. Influence of exotic earthworm invasion on soil organic matter, microbial biomass and denitrification potential in forest soils of the northeastern United States. *Applied Soil Ecology* 9, 197–202.
- Calcagno, V., de Mazancourt, C., 2010. *glmulti*: an R package for easy automated model selection with (generalized) linear models. *Journal of Statistical Software* 34, 1–29, 2010.
- Castle, S.C., Song, Z., Gohl, D.M., Gutknecht, J.L.M., Rosen, C.J., Sadowsky, M.J., Samac, D.A., Kinkel, L.L., 2018. DNA template dilution impacts amplicon sequencing-based estimates of soil fungal diversity. *Phytophysics Journal* 2, 100–107.
- Craven, D., Thakur, M.P., Cameron, E.K., Frelich, L.E., Beauséjour, R., Blair, R.B., Blossley, B., Burtis, J., Choi, A., Dávalos, A., Fahey, T.J., Fisichelli, N.A., Gibson, K., Handa, I.T., Hopfensperger, K., Loss, S.R., Nuzzo, V., Maerz, J.C., Sackett, T., Scharenbroch, B.C., Smith, S.M., Vellend, M., Umek, L.G., Eisenhauer, N., 2017. The unseen invaders: introduced earthworms as drivers of change in plant communities in North American forests (a meta-analysis). *Global Change Biology* 23, 1065–1074.
- Crooks, J.A., 2002. Characterizing ecosystem-level consequences of biological invasions: the role of ecosystem engineers. *Oikos* 97, 153–166.
- de Menezes, A.B., Prendergast-Miller, M.T., Macdonald, L.M., Toscani, P., Baker, G., Farrell, M., Wark, T., Richardson, A.E., Thrall, P.H., 2018. Earthworm-induced shifts in microbial diversity in soils with rare versus established invasive earthworm populations. *FEMS Microbiology Ecology* 94.
- Dempsey, M.A., Fisk, M.C., Fahey, T.J., 2011. Earthworms increase the ratio of bacteria to fungi in northern hardwood forest soils, primarily by eliminating the organic horizon. *Soil Biology and Biochemistry* 43, 2135–2141.
- Eilers, K.G., Debenport, S., Anderson, S., Fierer, N., 2012. Digging deeper to find unique microbial communities: the strong effect of depth on the structure of bacterial and archaeal communities in soil. *Soil Biology and Biochemistry* 50, 58–65.
- Eisenhauer, N., Partsch, S., Parkinson, D., Scheu, S., 2007. Invasion of a deciduous forest by earthworms: changes in soil chemistry, microflora, microarthropods and vegetation. *Soil Biology and Biochemistry* 39, 1099–1110.
- Ferlian, O., Eisenhauer, N., Aguirrebengoa, M., Camara, M., Ramirez-Rojas, I., Santos, F., Tanalgo, K., Thakur, M.P., 2018. Invasive earthworms erode soil biodiversity: a meta-analysis. *Journal of Animal Ecology* 87, 162–172.
- Fischer, O.A., Matlova, L., Bartl, J., Dvorska, L., Svastova, P., du Maine, R., Melicharek, I., Bartos, M., Pavlik, I., 2003. Earthworms (*Oligochaeta*, *Lumbricidae*) and mycobacteria. *Veterinary Microbiology* 91, 325–338.
- Frelich, L.E., Blossley, B., Cameron, E.K., Dávalos, A., Eisenhauer, N., Fahey, T., Ferlian, O., Groffman, P.M., Larson, E., Loss, S.R., Maerz, J.C., Nuzzo, V., Yoo, K., Reich, P.B., 2019. Side-swiped: ecological cascades emanating from earthworm invasions. *Frontiers in Ecology and the Environment* 17, 502–510.
- Gohl, D.M., Vangay, P., Garbe, J., MacLean, A., Hauge, A., Becker, A., Gould, T.J., Clayton, J.B., Johnson, T.J., Hunter, R., Knights, D., Beckman, K.B., 2016. Systematic improvement of amplicon marker gene methods for increased accuracy in microbiome studies. *Nature Biotechnology* 34, 942–949.
- Hale, C.M., Frelich, L.E., Reich, P.B., 2005a. Exotic European earthworm invasion dynamics in northern hardwood forests of Minnesota, USA. *Ecological Applications* 15, 848–860.
- Hale, C.M., Frelich, L.E., Reich, P.B., 2006. Changes in hardwood forest understory plant communities in response to European earthworm invasions. *Ecology* 87, 1637–1649.
- Hale, C.M., Frelich, L.E., Reich, P.B., Pastor, J., 2005b. Effects of European earthworm invasion on soil characteristics in northern hardwood forests of Minnesota. *Ecosystems* 8, 911–927. USA.
- Hendrix, P.F., Mac, A., Callahan, J., Drake, John M., Huang, C.-Y., James, S.W., Snyder, B.A., Zhang, W., 2008. Pandora's box contained bait: the global problem of introduced earthworms. *Annual Review of Ecology, Evolution, and Systematics* 39, 593–613.
- Higgins, S.A., Schadt, C.W., Matheny, P.B., Löffler, F.E., 2018. Phylogenomics reveal the dynamic evolution of fungal nitric oxide reductases and their relationship to secondary metabolism. *Genome Biology and Evolution* 10, 2474–2489.
- Hites, R.A., 2019. Correcting for censored environmental measurements. *Environmental Science & Technology* 53, 11059–11060.
- Hoeftner, K., Monard, C., Santonja, M., Cluzeau, D., 2018. Feeding behaviour of epigeic earthworm species and their impacts on soil microbial communities. *Soil Biology and Biochemistry* 125, 1–9.
- Horn, M.A., Schramm, A., Drake, H.L., 2003. The earthworm gut: an ideal habitat for ingested N_2O -producing microorganisms. *Applied and Environmental Microbiology* 69, 1662–1669.
- Ishii, S., 2020. Quantification of antibiotic resistance genes for environmental monitoring: current methods and future directions. *Current Opinion in Environmental Science & Health* 16, 47–53.
- Ishii, S., Segawa, T., Okabe, S., 2013. Simultaneous quantification of multiple food- and waterborne pathogens by use of microfluidic quantitative PCR. *Applied and Environmental Microbiology* 79, 2891–2898.
- Ishii, S., Song, Y., Rathnayake, L., Tumendelger, A., Satoh, H., Toyoda, S., Yoshida, N., Okabe, S., 2014. Identification of key nitrous oxide production pathways in aerobic partial nitrifying granules. *Environmental Microbiology* 16, 3168–3180.
- Isobe, K., Oka, H., Watanabe, T., Tateno, R., Urakawa, R., Liang, C., Senoo, K., Shibata, H., 2018. High soil microbial activity in the winter season enhances nitrogen cycling in a cool-temperate deciduous forest. *Soil Biology and Biochemistry* 124, 90–100.

- James, S.W., Hendrix, P.F., 2004. Invasion of exotic earthworms into north America and other regions. In: Edwards, C.A. (Ed.), *Earthworm Ecology*. CRC Press, Boca Raton, Florida, pp. 75–88.
- Karsten, G.R., Drake, H.L., 1997. Denitrifying bacteria in the earthworm gastrointestinal tract and in vivo emission of nitrous oxide (N₂O) by earthworms. *Applied and Environmental Microbiology* 63, 1878–1882.
- Köljal, U., Nilsson, R.H., Abarenkov, K., Tedersoo, L., Taylor, A.F.S., Bahram, M., Bates, S.T., Bruns, T.D., Bengtsson-Palme, J., Callaghan, T.M., Douglas, B., Drenkhan, T., Eberhardt, U., Dueñas, M., Grebenc, T., Griffith, G.W., Hartmann, M., Kirk, P.M., Kohout, P., Larsson, E., Lindahl, B.D., Lücking, R., Martín, M.P., Matheny, P.B., Nguyen, N.H., Niskanen, T., Oja, J., Peay, K.G., Peintner, U., Peterson, M., Pöldmaa, K., Saag, L., Saar, I., Schüssler, A., Scott, J.A., Senés, C., Smith, M.E., Suija, A., Taylor, D.L., Telleria, M.T., Weiss, M., Larsson, K.-H., 2013. Towards a unified paradigm for sequence-based identification of fungi. *Molecular Ecology* 22, 5271–5277.
- Larson, E.R., Kipfmüller, K.F., Hale, C.M., Frelich, L.E., Reich, P.B., 2010. Tree rings detect earthworm invasions and their effects in northern hardwood forests. *Biological Invasions* 12, 1053–1066.
- Li, C., Hu, H.-W., Chen, Q.-L., Chen, D., He, J.-Z., 2020. Niche differentiation of clade A comammox *Nitrospira* and canonical ammonia oxidizers in selected forest soils. *Soil Biology and Biochemistry* 149, 107925.
- Lladó, S., López-Mondéjar, R., Baldrian, P., 2017. Forest soil bacteria: diversity, involvement in ecosystem processes, and response to global change. *Microbiology and Molecular Biology Reviews* 81, e00063, 00016.
- López-Mondéjar, R., Voríšková, J., Větrovský, T., Baldrian, P., 2015. The bacterial community inhabiting temperate deciduous forests is vertically stratified and undergoes seasonal dynamics. *Soil Biology and Biochemistry* 87, 43–50.
- Love, M.I., Huber, W., Anders, S., 2014. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* 15, 550.
- Lyttle, A., Yoo, K., Hale, C., Aufdenkampe, A., Sebestyen, S., 2011. Carbon–mineral interactions along an earthworm invasion gradient at a sugar maple forest in northern Minnesota. *Applied Geochemistry* 26, S85–S88.
- Lyttle, A., Yoo, K., Hale, C., Aufdenkampe, A., Sebestyen, S.D., Resner, K., Blum, A., 2015. Impact of exotic earthworms on organic carbon sorption on mineral surfaces and soil carbon inventories in a northern hardwood forest. *Ecosystems* 18, 16–29.
- Martin, F., Kohler, A., Murat, C., Veneault-Fourrey, C., Hibbett, D.S., 2016. Unearthing the roots of ectomycorrhizal symbioses. *Nature Reviews Microbiology* 14, 760–773.
- McDonald, D., Price, M.N., Goodrich, J., Nawrocki, E.P., DeSantis, T.Z., Probst, A., Andersen, G.L., Knight, R., Hugenholtz, P., 2012. An improved greengenes taxonomy with explicit ranks for ecological and evolutionary analyses of Bacteria and Archaea. *The ISME Journal* 6, 610–618.
- McMurdie, P.J., Holmes, S., 2013. *phyloseq*: an R package for reproducible interactive analysis and graphics of microbiome census data. *PLoS One* 8, e61217.
- Medina-Sauza, R.M., Álvarez-Jiménez, M., Delhal, A., Reverchon, F., Blouin, M., Guerrero-Analco, J.A., Cerdán, C.R., Guevara, R., Villain, L., Barois, I., 2019. Earthworms building up soil microbiota, a review. *Frontiers in Environmental Science* 7.
- Nguyen, N.H., Song, Z., Bates, S.T., Branco, S., Tedersoo, L., Menke, J., Schilling, J.S., Kennedy, P.G., 2016. FUNGuild: an open annotation tool for parsing fungal community datasets by ecological guild. *Fungal Ecology* 20, 241–248.
- Oksanen, J., Blanchet, F.G., Friendly, M., Kindt, R., Legendre, P., McGinn, D., Minchin, P.R., O'Hara, R.B., Simpson, G.L., Solymos, P., Stevens, M.H.H., Szoecs, E., Wagner, H., 2019. *vegan*: Community Ecology Package, R package version 2.5-6. <http://CRAN.R-project.org/package=vegan>.
- Osburn, E.D., Barrett, J.E., 2020. Abundance and functional importance of complete ammonia-oxidizing bacteria (comammox) versus canonical nitrifiers in temperate forest soils. *Soil Biology and Biochemistry* 145, 107801.
- Oshiki, M., Segawa, T., Ishii, S., 2018. Nitrogen cycle evaluation (NICE) chip for the simultaneous analysis of multiple N-cycle associated genes. *Applied and Environmental Microbiology* 84, e02615-17.
- Pass, D.A., Morgan, A.J., Read, D.S., Field, D., Weightman, A.J., Kille, P., 2015. The effect of anthropogenic arsenic contamination on the earthworm microbiome. *Environmental Microbiology* 17, 1884–1896.
- Phillips, H.R.P., Guerra, C.A., Bartz, M.L.C., Briones, M.J.I., Brown, G., Crowther, T.W., Ferlian, O., Gongalsky, K.B., van den Hoogen, J., Krebs, J., Orgiazzi, A., Routh, D., Schwarz, B., Bach, E.M., Bennett, J.M., Brose, U., Decaens, T., König-Ries, B., Loreau, M., Mathieu, J., Mulder, C., van der Putten, W.H., Ramirez, K.S., Rillig, M.C., Russell, D., Rutgers, M., Thakur, M.P., de Vries, F.T., Wall, D.H., Wardle, D.A., Arai, M., Ayuke, F.O., Baker, G.H., Beauséjour, R., Bedano, J.C., Birkhofer, K., Blanchart, E., Blossey, B., Bolger, T., Bradley, R.L., Callahan, M.A., Capowiez, Y., Caulfield, M.E., Choi, A., Crotty, F.V., Crumsey, J.M., Dávalos, A., Díaz Cosin, D.J., Domínguez, A., Duhour, A.E., van Eekeren, N., Emmerling, C., Falco, L.B., Fernández, R., Fonte, S.J., Fragoso, C., Franco, A.L.C., Fugère, M., Fusilero, A.T., Gholami, S., Gundale, M.J., López, M.G., Hackenberger, D.K., Hernández, L.M., Hishi, T., Holdsworth, A.R., Holmström, M., Hopfensperger, K.N., Lwanga, E.H., Huhta, V., Hurisso, T.T., Iannone, B.V., Iordache, M., Joschko, M., Kaneko, N., Kanianska, R., Keith, A.M., Kelly, C.A., Kernecker, M.L., Klaminder, J., Koné, A.W., Kooch, Y., Kukkonen, S.T., Lalthanzara, H., Lammel, D.R., Lebedev, I.M., Li, Y., Jesus Lidon, J.B., Lincoln, N.K., Loss, S.R., Marichal, R., Matula, R., Moos, J.H., Moreno, G., Morón-Ríos, A., Muys, B., Neirynck, J., Norgrove, L., Novo, M., Nuutinen, V., Nuzzo, V., Pansu, J., Paudel, S., Pères, G., Pérez-Camacho, L., Piñeiro, R., Ponge, J.-F., Rashid, M.I., Rebollo, S., Rodeiro-Iglesias, J., Rodríguez, M.A., Roth, A.M., Rousseau, G.X., Rozen, A., Sayad, E., van Schaik, L., Scharenbroch, B.C., Schirrmann, M., Schmidt, O., Schröder, B., Seiber, J., Shashkov, M.P., Singh, J., Smith, S.M., Steinwandter, M., Talavera, J.A., Trigo, D., Tsukamoto, J., de Valença, A.W., Vanek, S.J., Virto, I., Wackett, A.A., Warren, M.W., Wehr, N.H., Whalen, J.K., Wironen, M.B., Wolters, V., Zenkova, I.V., Zhang, W., Cameron, E.K., Eisenhauer, N., 2019. Global distribution of earthworm diversity. *Science* 366, 480–485.
- Resner, K., Yoo, K., Sebestyen, S.D., Aufdenkampe, A., Hale, C., Lyttle, A., Blum, A., 2015. Invasive earthworms deplete key soil inorganic nutrients (Ca, Mg, K, and P) in a northern hardwood forest. *Ecosystems* 18, 89–102.
- Saltmarsh, D.M., Bowser, M.L., Morton, J.M., Lang, S., Shain, D., Dial, R., 2016. Distribution and abundance of exotic earthworms within a boreal forest system in southcentral Alaska. *NeoBiota* 28.
- Scheu, S., Parkinson, D., 1994. Effects of earthworms on nutrient dynamics, carbon turnover and microorganisms in soils from cool temperate forests of the Canadian Rocky Mountains — laboratory studies. *Applied Soil Ecology* 1, 113–125.
- Sun, M., Chao, H., Zheng, X., Deng, S., Ye, M., Hu, F., 2020. Ecological role of earthworm intestinal bacteria in terrestrial environments: a review. *The Science of the Total Environment* 740, 140008.
- Svensson, B.H., Boström, U., Klemetson, L., 1986. Potential for higher rates of denitrification in earthworm casts than in the surrounding soil. *Biology and Fertility of Soils* 2, 147–149.
- Tang, Y., Yu, G., Zhang, X., Wang, Q., Ge, J., Liu, S., 2018. Changes in nitrogen-cycling microbial communities with depth in temperate and subtropical forest soils. *Applied Soil Ecology* 124, 218–228.
- Tedersoo, L., Pärtel, K., Jairus, T., Gates, G., Pöldmaa, K., Tamm, H., 2009. Ascomycetes associated with ectomycorrhizas: molecular diversity and ecology with particular reference to the Helotiales. *Environmental Microbiology* 11, 3166–3178.
- Thakuria, D., Schmidt, O., Finan, D., Egan, D., Doohan, F.M., 2010. Gut wall bacteria of earthworms: a natural selection process. *The ISME Journal* 4, 357–366.
- Tourna, M., Stieglmeier, M., Spang, A., Könneke, M., Schintlmeister, A., Urlich, T., Engel, M., Schlöter, M., Wagner, M., Richter, A., Schleper, C., 2011. *Nitrososphaera viennensis*, an ammonia oxidizing archaeon from soil. *Proceedings of the National Academy of Sciences of the United States of America* 108, 8420–8425.
- Tu, Q., Lin, L., Cheng, L., Deng, Y., He, Z., 2018. NCycDB: a curated integrative database for fast and accurate metagenomic profiling of nitrogen cycling genes. *Bioinformatics* 35, 1040–1048.
- Wackett, A.A., Yoo, K., Olofsson, J., Klaminder, J., 2018. Human-mediated introduction of geoeengineering earthworms in the Fennoscandian arctic. *Biological Invasions* 20, 1377–1386.
- Wang, Z., Cao, Y., Zhu-Barker, X., Nicol, G.W., Wright, A.L., Jia, Z., Jiang, X., 2019. Comammox *Nitrospira* clade B contributes to nitrification in soil. *Soil Biology and Biochemistry* 135, 392–395.